

Event-TimeRAF: AI Claim Verification Log

Verified against run 20260723T112033170131Z

Evidence contract. The executed verification notebook checks all 69 manifest artifacts by SHA-256, reruns saved predictors, reconstructs metrics from 172,776 test forecast points, reruns paired block-bootstrap ablations, and regenerates the manuscript figures. The immutable file `event_timeraf_final_run.zip` is the numerical source of truth.

Claim	Checked how	Result
Run artifacts are unchanged and attributable 48,332 windows; 34,020/7,113/7,199 chronological split Context/model dimensions are 86 and 152	Recomputed SHA-256 for all 69 files listed in the archived manifest	Confirmed
	Reconstructed saved NPZ arrays and joined window metadata; asserted shapes and split counts	Confirmed
	Counted saved feature names and inspected the M09 model: 23+14+9+39=85; 85+51+6+9=151	Corrected
M04 reports MSE/MAE/RMSE/ R^2 of 26.185/3.125/5.117/0.379 Full M09 improves on M04	Executed archived M04 on the reconstructed test matrix and recomputed all four metrics	Confirmed
	Executed M09; MSE is 26.712 versus 26.185 for M04; paired M09–M04 interval is [−0.026, 1.182]	Corrected
Retrieval improves frozen Chronos-Bolt conclusively	Rebuilt validation-selected fusion; MSE changes 28.941 to 28.709, but paired interval [−0.665, 0.152] crosses zero	Corrected
Component-ablation values are measured	Reran eight prediction-level comparisons with 500-resample, 24-hour paired block bootstrap	Confirmed
M12 drift gating is a verified final result	Checked manifest membership and hashes; M12 appears only in post-manifest edited artifacts and was removed from the paper	Corrected
Seeded random retrieval is bitwise portable	Compared a NumPy 2.4.1 replay with NumPy 2.0.2 evidence; M05/A01 are reconstructed from checksum-verified evidence	Flagged unverified
Storm-event context is available in strict real time	Inspected NOAA audit fields; publication timestamps are absent, so event results are retrospective sensitivity analyses	Flagged unverified
Findings generalize geographically and runtime is hardware-normalized	Checked dataset inventory and manifest hardware fields; only Los Angeles County is evaluated and hardware identifiers are absent	Flagged unverified

AI writing removed or corrected (before → after).

Dimensions	“86-dimensional context; 40 event features; 152 inputs”	“85-dimensional context; 39 event features; 151 per-horizon inputs”
Headline result	“M12 is the best final model”	M12 result removed; M04 is the strongest manifest-backed model
Drift method	Generic weighted drift-score equation	Five implemented raw statistics, validation median/MAD scaling, clipping, and the 0.90 validation quantile
Claim scope	Event-aware findings phrased as operational evidence	Retrospective event sensitivity, one-county validation, and non-normalized runtime stated explicitly